Testing Life Cycle,Principles of Software Testing,Types of Software Testing

Objectives of Software Testing

- Find defects and prevent defects
- Make sure the software is good in quality
- Meet product requirements
- Fit for use

Software Testing Life Cycle

1. **Requirement Analysis:** Understand what needs to be tested.
2. **Test Planning:** Define the strategy, scope, and timeline.
3. **Test Case Development:** Write specific steps and expected outcomes.
4. **Environment Setup:** Prepare the hardware and software platforms.
5. **Test Execution:** Run the tests and record results.
6. **Defect Logging:** Report and track found bugs.
7. **Test Closure:** Finalize reports and confirm completion.

Seven Principles of Software Testing

1. Early testing
2. Testing shows presence of defects
3. Exhaustive testing is not possible
4. Defect clustering
5. Pesticide Paradox
6. Testing is context dependent
7. Absence of error fallacy

Software Development Lifecycle

1. Requirement Gathering and Analysis
2. Design
3. Development
4. Testing
5. Deploy to production
6. Maintenance

Types of Software Testing

Functional (testing the functionality of a particular software)
- Unit test, Integration test, System test, User acceptance test, regression test, sanity test, smoke test, usability test

- Nonfunctional (test the performance of the software such as response time, latency, throughput)
- Performance testing, load testing, stress testin